

Scarlett Enos

CS 499

CS 330 Artifact Narrative

This artifact is from the CS330 final project where the goal was to create a scene with complex objects and textures. I created it in February 2026. The scene has two coffee mugs on coasters sitting on a marble countertop with an open window in the background lighting the scene. I decided to include this artifact in my e-portfolio because I identified several areas where I could improve my scene and I also enjoyed working on this project during CS330. This project was meant to demonstrate my understanding of computational graphics and how to use that understanding to render objects, lighting, and textures. For improvements, I mainly focused on the SceneManager.cpp file. The file was improved by cleaning up a lot of repeated code and making the code overall easier to maintain and update. In the original version, many of the textures, materials, and rendering steps were written manually over and over again. To fix the repetition, I implemented helper methods to make the creation of materials more organized. I also fixed repetition in the texture loading section by organizing the textures into structures and loops instead of manually loading each texture one at a time.

I believe I met the course outcomes that I originally planned to focus on in Module One because I improved the overall organization and structure of my code as well as improving my documentation by adding more thorough comments explaining what I am doing in my code. An update to my outcome coverage plan is when I was trying to run my code, I noticed that it was missing a critical file that was needed for the project to fully compile, specifically the ShaderManager.h file. I think when I initially uploaded it to GitHub for submission during the

CS330 course, I somehow managed to delete that file. Since that file was missing from my GitHub and I no longer have access to the course, I could not rebuild and test my artifact with all my enhancements. Because of that, I plan to recreate the shader file and continue to troubleshoot problems throughout the rest of the course so the project can become fully functional and run again. This was the biggest challenge I came across during my enhancement of this artifact because I spent all week sorting through old versions of the project on my computer to hopefully locate the missing file to no success. Through my enhancements, I learned how important organization is and how large repetitive sections of code can make maintenance and updates harder as a project grows more complex. I also learned how important it is to save previous versions of your project in case you need to recover files later.

References I used to make my enhancements

Data structures. (n.d.). cplusplus.com. Retrieved May 22, 2026, from

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